

DeepSeek-V2

Efficient MoE Language Model
with Multi-Head Latent Attention

236B

Total Parameters

21B

Activated per Token

128K

Context Window

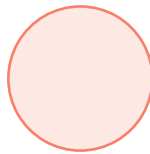
A Strong, Economical, and Efficient Language Model · DeepSeek AI (2024)

The LLM Scaling Dilemma: Performance vs. Training Cost and Inference Speed



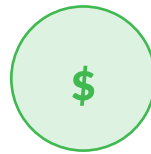
Larger Models = Better Intelligence

Dense scaling laws demand proportionally more parameters. State-of-the-art models exceed 70B parameters — every gain in quality multiplies training and inference cost.



KV Cache Bottleneck

Multi-Head Attention caches $O(n \cdot h \cdot d)$ key/value vectors per layer. At 128K context windows this becomes the dominant memory constraint, throttling generation throughput.

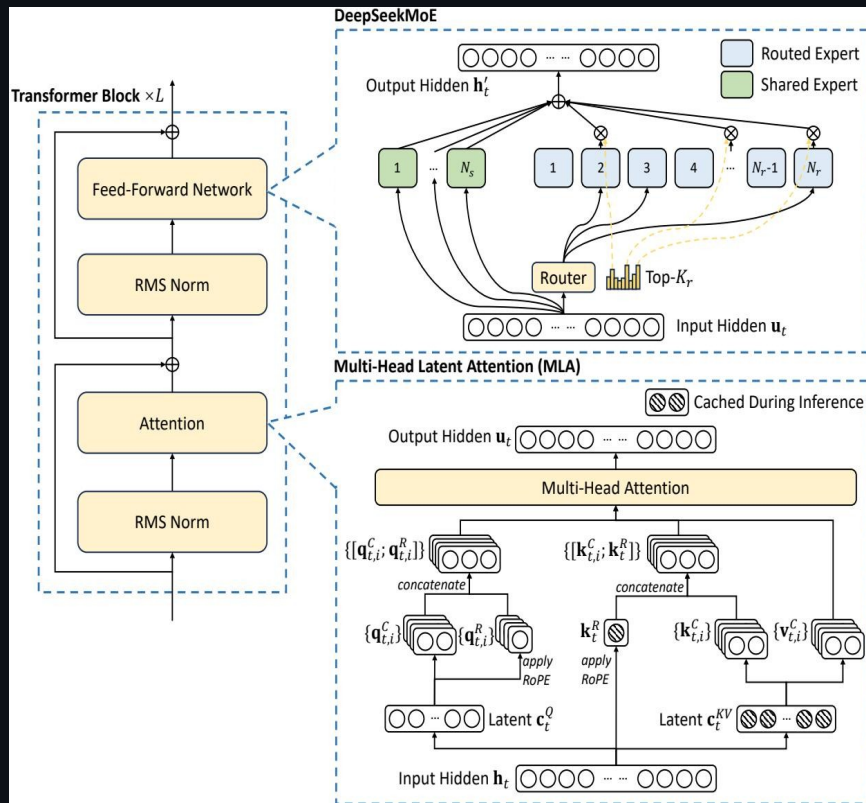


Prohibitive Training Cost

Training a dense 67B model is already expensive. Scaling without architectural efficiency gains is economically infeasible — and wasteful if most parameters aren't needed.

Solution: DeepSeek-V2 addresses all three — 42.5% lower training cost, 93.3% KV cache reduction, 5.76× generation throughput

DeepSeek-V2 Architecture: MLA + DeepSeekMoE in Each Transformer Block



MLA — Attention Layer

Replaces standard MHA with low-rank key-value joint compression. A compact latent vector is cached per token; full K/V reconstructed on-the-fly.

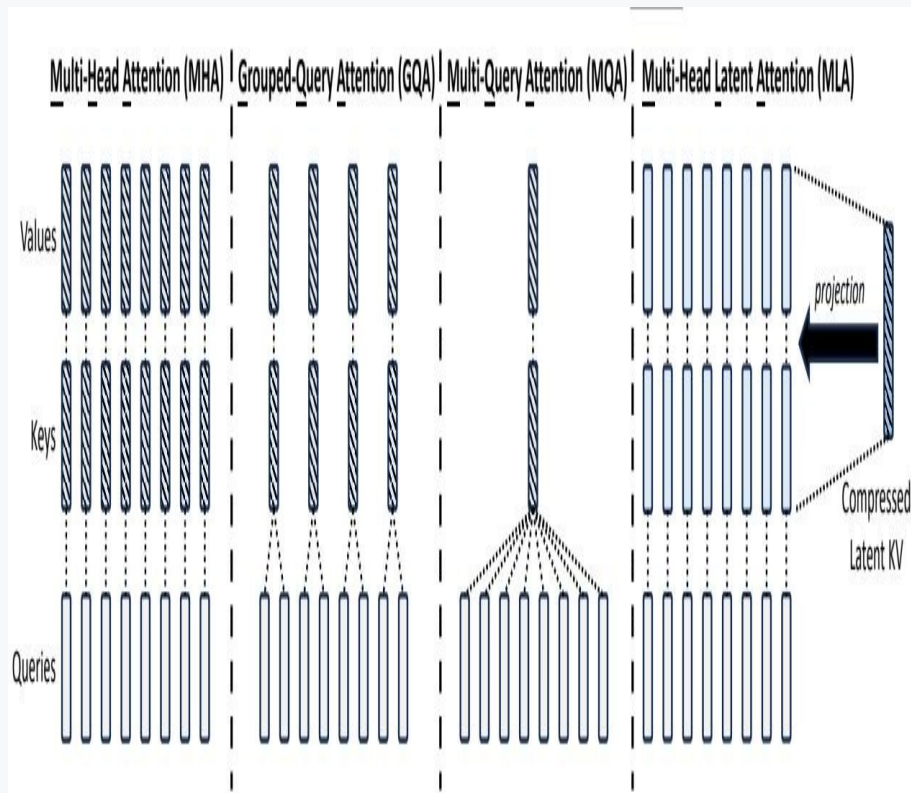
DeepSeekMoE — FFN Layer

Replaces the dense FFN with fine-grained shared + sparsely routed experts. Only 21B of 236B parameters activate per token.

Scale & Training

236B total params · 21B activated · 128K context · Pretrained on 8.1T tokens from a multi-source corpus.

Multi-Head Latent Attention: 93.3% KV Cache Compression



Standard MHA

Caches full K and V vectors for every attention head per token. At long contexts this becomes a severe memory bottleneck limiting batch size and throughput.

GQA / MQA

Reduce cache by sharing or merging heads across groups. Lowers memory, but sacrifices model quality compared to full MHA — a tradeoff MLA avoids.

MLA (DeepSeek-V2)

Low-rank joint KV compression: caches one compact latent vector c_{KV} per token. Full K/V reconstructed via learned projections. Decoupled RoPE. Outperforms MHA with 93.3% less cache.

Result: MLA outperforms MHA with 93.3% less KV cache

DeepSeekMoE: Fine-Grained Experts with Shared Expert Isolation

Shared Experts (N_s)

Always activated — capture universal knowledge common across all tokens and tasks, preventing expert redundancy in routed layers.

Device-Limited Routing

Each token is constrained to experts residing on at most M devices, drastically reducing expensive cross-node communication during training and inference.

Routed Experts (N_r)

Sparsely activated via top- K_r router. Fine-grained expert segmentation increases the potential for specialization per knowledge domain.

Load Balancing & Overflow

Auxiliary losses enforce balanced utilization across experts and devices. A token-dropping strategy during training prevents expert overflow without affecting inference quality.

Result: 236B total parameters, only 21B activated — training costs **42.5% lower** than a 67B dense predecessor model.

42.5% Training Cost Reduction and 5.76× Generation Throughput vs. DeepSeek 67B

DeepSeek 67B (Dense Baseline)

Relative metrics vs. DeepSeek 67B · no absolute values fabricated

Training Cost

Baseline

KV Cache Size

Baseline

Max Generation Throughput

1×

DeepSeek-V2 (MoE)

Training Cost

-42.5%

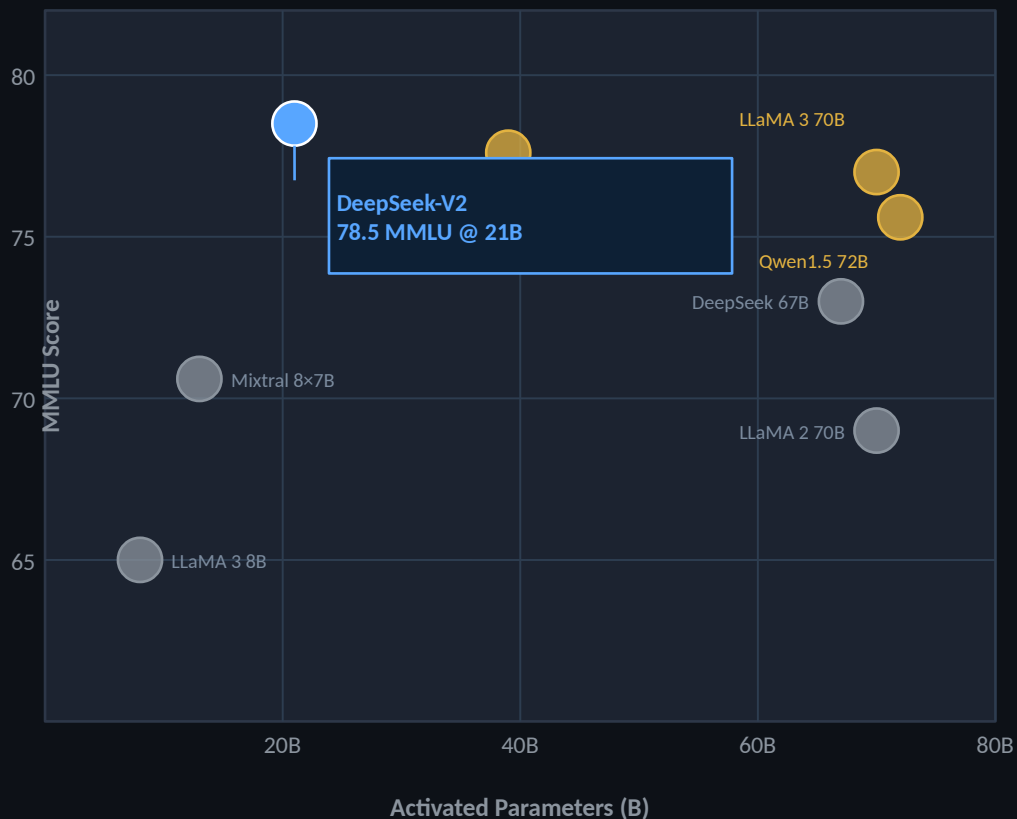
KV Cache Size

-93.3%

Max Generation Throughput

5.76×

DeepSeek-V2 Tops MMLU at Only 21B Activated Parameters



Model	Act. (B)	MMLU
DeepSeek-V2	21	78.5
LLaMA 3 70B	70	77.0
Mixtral 8×22B	39	77.6
Qwen1.5 72B	72	75.6
DeepSeek 67B	67	73.0
Mixtral 8×7B	13	70.6
LLaMA 2 70B	70	69.0
LLaMA 3 8B	8	65.0

DeepSeek-V2 Chat (RL) Rivals Closed-Source Models on Chat Benchmarks

Alignment Pipeline: Pretrain (8.1T tokens) → SFT (1.5M sessions) → GRPO Reinforcement Learning

AlpacaEval 2.0

Length-Controlled Win Rate

38.9%

Open-ended English instruction following evaluated by GPT-4 as judge. Length-controlled win rate eliminates verbosity bias. DeepSeek-V2 Chat (RL) is competitive with leading closed-source systems.

MT-Bench

Score out of 10

8.97

Multi-turn conversation benchmark covering writing, coding, reasoning, and STEM. Score of 8.97 places DeepSeek-V2 Chat (RL) among the top open-source models across all task categories.

AlignBench

Overall Score (Chinese)

7.91

Comprehensive Chinese alignment benchmark. DeepSeek-V2 Chat (RL) achieves #1 among all open-source models and outperforms most closed-source Chinese systems, demonstrating strong multilingual alignment.

Key Takeaways

Architectural Innovation Unlocks Efficient Scaling

01 MLA: Strictly Better Than MHA

Low-rank KV joint compression cuts cache by 93.3% while exceeding standard MHA performance — a superior replacement for GQA/MQA.

02 DeepSeekMoE: Economical 236B Model

Shared + fine-grained routed experts enable training 236B parameters at 42.5% lower cost than a 67B dense model.

03 Pareto-Optimal on MMLU

21B activated params match or exceed 70B-class dense models (LLaMA 3 70B, Qwen1.5 72B) — more capability per FLOP.

04 5.76× Generation Throughput

Compressed KV cache enables near-6× higher generation throughput over DeepSeek 67B. Smaller per-token memory footprint allows larger batch sizes and higher GPU utilization at serving time.

05 Strongest Open-Source Chinese Model

Chat (RL) tops AlignBench (7.91), MT-Bench (8.97), and AlpacaEval 2.0 (38.9% win rate), leading all open-source and most closed-source models in Chinese.